World Knowledge as a Disambiguation Signal in Turkish Parsing: Human Use, Model Underuse

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Background. Prior work on Turkish relative-clause (RC) attachment has examined how comprehenders resolve the ambiguity by manipulating factors such as relatively broad pragmatic possibility contrasts, as illustrated in (1) (Akal 2021; Dinçtopal-Deniz 2010; Kırkıcı 2004). Building on this literature, we isolate pragmatic plausibility confirmed by norming ratings (2) more narrowly as opposed to pragmatic possibility (1) by holding the syntactic environment constant while selectively biasing the plausibility of attachment to the higher vs. lower nominal. In (2), for instance, both *model* and *bodyguard* are structurally and more broadly pragmatically possible RC hosts; however, world knowledge (WK) renders *model* a far more plausible agent of *run away* than *bodyguard*, thereby isolating a targeted plausibility cue that bears directly on attachment choice.

Human Experiment. To test whether such pragmatic plausibility as in (2) modulates RC attachment in Turkish, we conducted an online speeded forced-choice study on Ibex Farm with 86 native Turkish speakers using 40 critical items (20 High-WK, 20 Low-WK), where participants answered a *who*-question selecting the intended RC host (N1 vs N2). Lexical frequency was controlled by checking corpus-based frequencies for the critical nouns and verbs and avoiding systematic frequency differences across conditions in addition to animacy. In Low-WK contexts, where plausibility favors attaching the RC to the lower noun, participants chose high attachment (HA) only 26.3 % of the time. In High-WK contexts, where plausibility favors HA, they chose HA 65.2 % of the time, a 5.2× odds increase ($\beta_{(}WK_{)} = 1.65 \pm 0.11$, $z = 14.75$, $p < 10^{-50}$). Without plausibility cues, the baseline HA rate was 45.7 %, confirming a modest low attachment (LA) bias in Turkish (Akal 2021; Dinçtopal-Deniz 2010). A χ^2 test ($\chi^2_{(}1_{)} = 229.1$, $p = 1.7 \times 10^{-53}$) and item-wise Spearman correlation ($\rho = 0.85$, $p = 2 \times 10^{-6}$) corroborate that WK reliably drives attachment decisions.

LLM Experiments. Parallel forced-choice tests were administered to two LLMs on the same 40 critical items, using log probability (3) as a measure. Given the sentence context, we compared the model's mean per-token conditional log-probability for two short, syntactically licensed completions corresponding to the two attachment parses matched for length and selected the higher-probability completion as the model's predicted attachment. Turkish GPT-2 (ytu-ce-cosmos/turkish-gpt2) showed a stable low-attachment preference across conditions (High-WK: HA = 6/20 = 30% vs. LA = 14/20 = 70%; Low-WK: HA = 6/20 = 30% vs. LA = 14/20 = 70%), and neither condition deviated from chance in HA selection (two-sided binomial tests: $p = 0.115$in each). DeepSeek-R1-Distill-Qwen-1.5B-Turkish showed no reliable attachment preference overall (High-WK: HA = 12/20 = 60%, $p = 0.503$; Low-WK: HA = 10/20 = 50%, $p = 1.000$). Thus, unlike human readers—who integrate world knowledge to shift attachment—these models either maintain a stable LA-leaning pattern (70%) across conditions (Turkish GPT-2) or fail to show a reliable preference (DeepSeek-R1-Distill-Qwen-1.5B-Turkish), suggesting that pretraining and distillation alone do not induce robust world-knowledge integration in parsing.

Discussion. Mahowald et al. (2024) argue that LLMs can encode substantial world knowledge yet deploy unreliably. Our Turkish RC-attachment results sharpen this point: even when the relevant plausibility relation is simple and local, the models fail to consistently recruit it to resolve hierarchical attachment, consistent with a dissociation between fluent form and world-grounded interpretive control. Theoretically, these findings show that Turkish comprehenders integrate lexical semantics and world knowledge during RC attachment resolution, extending "semantic override" accounts beyond Indo-European cases (MacDonald et al. 1994; Trueswell et al. 1994). By contrast, contemporary LLMs maintain an unyielding LA or chance-level bias, failing to mirror the human pattern, consistent with recent cross-linguistic benchmarks (Cai et al. 2024; Lee et al. 2024; Karakaş

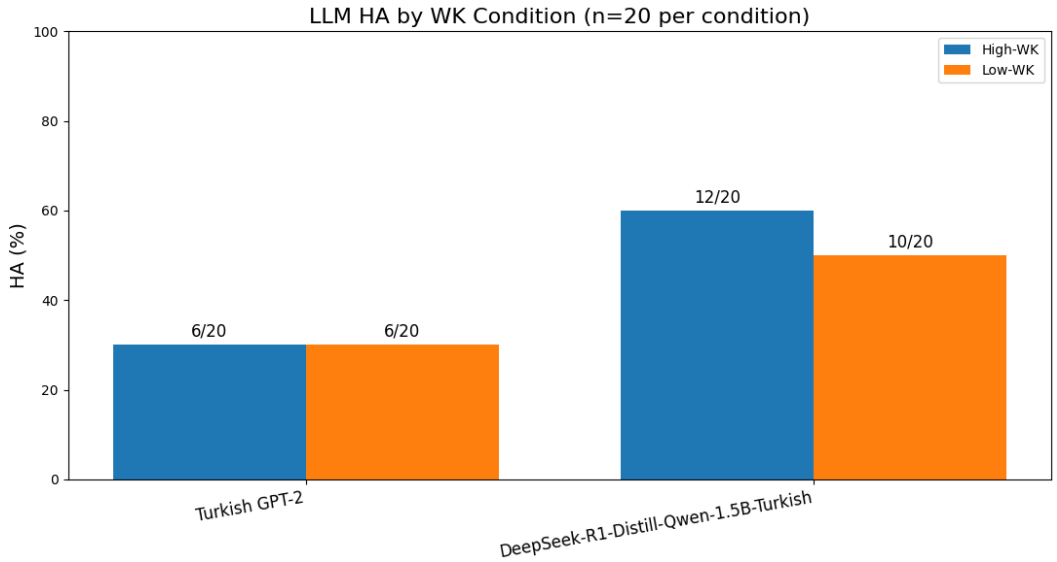
2026). In sum, our research (1) broadens empirical coverage by including Turkish RC attachment data, (2) highlights the universal role of real-world plausibility in parsing across typologically diverse grammars, and (3) provides a benchmark for future psycholinguistic and computational investigations at the interface of linguistic diversity and processing.

(1) Ev-i sil-en bebeğ-in anne-si duş al-dı.
Home-acc clean-REL baby-GEN mother-POSS shower take-PAST
'The mother of the baby, who cleaned the house took a shower.'

(2) Gazeteci-ler-den kaç-an manken-in koruma-sı güçlü-ydü
reporter-PL-ABL run-REL model-GEN bodyguard-POSS strong-PAST
'The bodyguard of the model, who ran away from reporters, was very strong.'

(3) $\overline{\log P}\,(\,y \mid x\,) = \frac{1}{k}\sum_{t=1}^{k} \log P\,(\,y_t \mid x, y_{<t}\,)$

(4)



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